Native Association

Confidence-Aware Human Perception in the Wild with a Foundation VLM

Igal Dmitriev ▪ Ofir Liba WSC Sports, Tel Aviv, Israel

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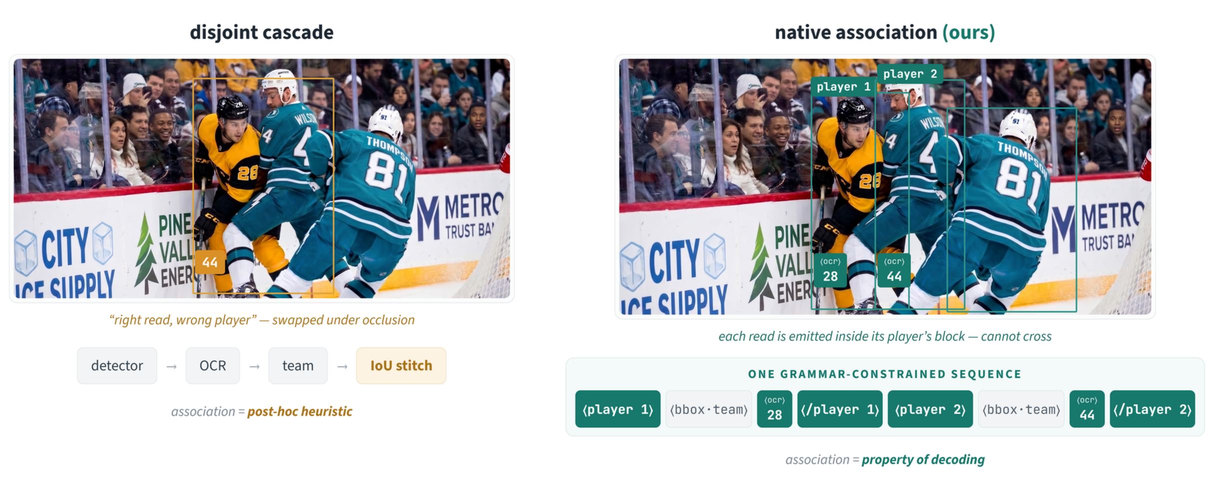
The problem: association

Reading *who is where, on which team, wearing which number* from a broadcast frame is normally done by **stitching** a detector, an OCR engine and classifiers together.

The stitch is the weak joint. Under occlusion it attaches a *correctly read* number to the *wrong* athlete. Even granted ground-truth boxes and one crop per athlete, cascades still misbind **15–18%** of the digits they read.



Left: a cascade reads "44" correctly and binds it to the wrong player. Right: each read is emitted *inside* its owner's <player> block — the binding step does not exist.

Three axes, kept separate

Syntactic validity — does the output parse into the schema on every frame?

Visual grounding — is every attribute anchored to a real, non-degenerate region?

Semantic association — is every attribute bound to the *correct* person?

Cascades fail chiefly the third. Frontier VLM APIs degrade on all three.

Data & protocol

Broadcast frames across **five sports** — hockey, soccer, gridiron football, tennis, basketball — LLM-pre-annotated and human-corrected in CVAT.

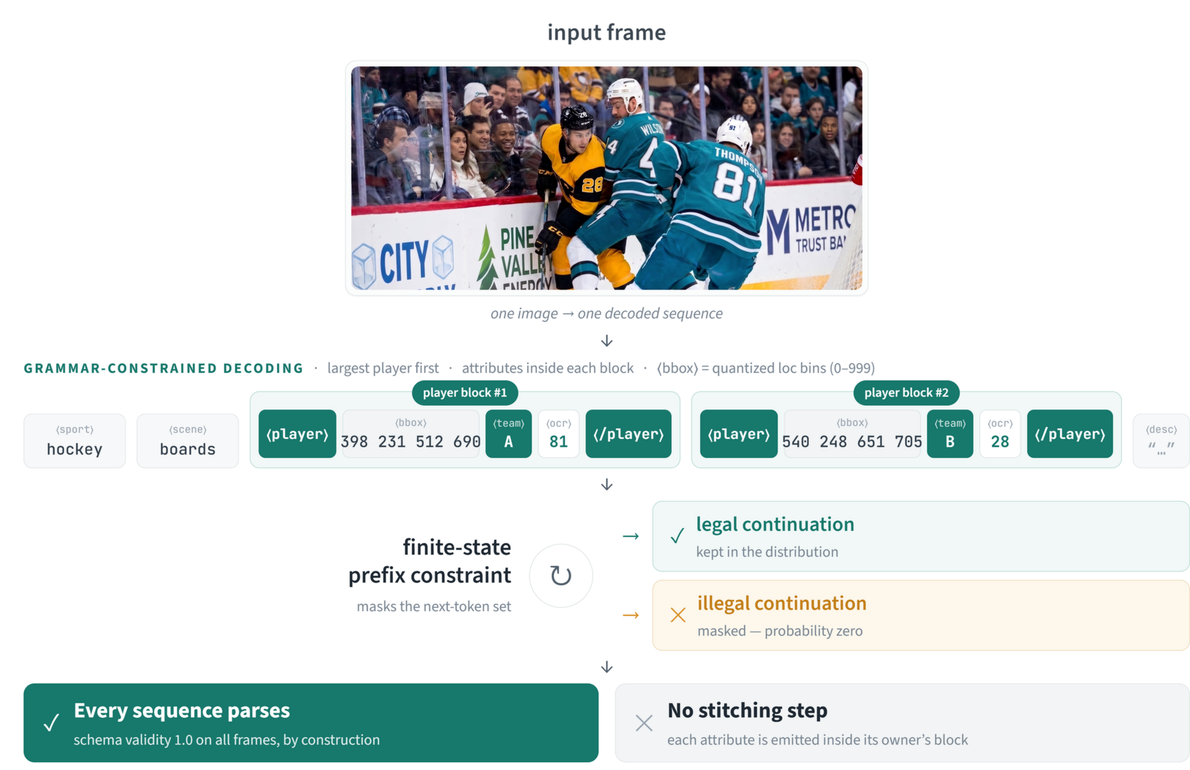
Frames (train/val/test)	4,584 / 577 / **567**
Players	12,999
OCR items	41,864

The test split is **frozen** and decoded once per variant, under a fingerprint guard: every evaluation recomputes the frame count, frame-set hash and annotation hash, and aborts on mismatch.

Source-level audit: 5,623/5,728 frames are unique-URL stills; only **4/567** test frames share a source video with training.

Method: one grammar-constrained pass

A single **0.77B** foundation VLM (Florence-2) is fine-tuned to emit *all* per-person attributes as **one sequence**, each attribute generated inside its owner's block. Both encoders frozen; decoder + shared token embeddings trained.



Two properties hold *by construction*, not by training:

1. Native association. Every OCR string is generated inside exactly one <player> block, so text–player binding is *read off the sequence* rather than reconstructed by a post-hoc IoU stitch.

2. Guaranteed validity. A prefix-allowed-tokens function admits only grammar-legal continuations, so every decode parses: **schema validity 1.0** on every frame, no repair, no reliance on the model having "learned" to be well-formed.

One replay ⇒ per-field confidence

One teacher-forced pass over the generated tokens (≈150 ms) recovers per-token probabilities. For a field F:

$$c(F) = \exp\Big(\tfrac{1}{k}\textstyle\sum_i \log P(y_{t_i} \mid y_{<t_i}, X)\Big)$$

The replay is *unmasked* — confidences are full-vocabulary beliefs, not renormalised over grammar-legal tokens.

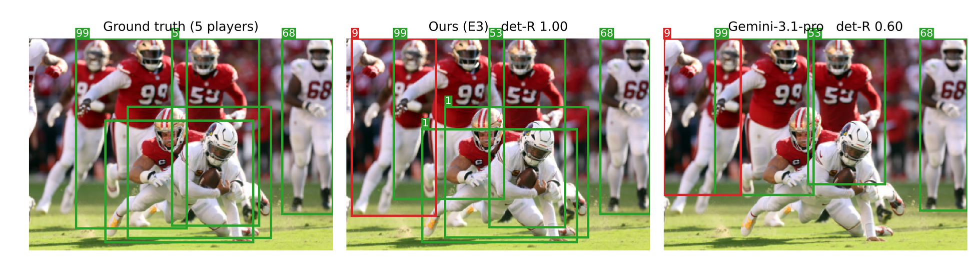
Results: frozen 567-frame multi-sport test

All systems scored by one suite, identical Hungarian matching at IoU $\geq$ 0.5.

Metric	**Ours** (E1b)	Gemini 2.5-pro	Gemini 3.1-pro
Detection F1	**0.950**	0.652	0.745
Match coverage	**0.950**	0.704	0.772
Jersey F1	**0.757**	0.704	0.737
OCR-text F1	**0.584**	0.476	0.502
AER (digits) ↓	**0.057**	0.242	0.214
Schema valid (%)	**100**	–	–
Latency (ms) ↓	**1224**	26009	8426

Grounding is the chasm. We detect at F1 **0.95** where the APIs reach **0.65–0.75**. Condition detection away and their reading deficit disappears — it is a *detection* deficit.

Association. Misbinding ≈ 4× lower than the APIs' (0.057 vs 0.21–0.24). **90–97%** of API misbindings are reads grounded to *no* athlete — the class in-block emission removes by construction.



Crowded frame: GT | ours | Gemini-3.1-pro. Detection recall 1.0 vs 0.6.

Stated honestly. Precision-by-abstention: baselines read only the easy players (coverage 0.70–0.77), which inflates their precision. We read everyone and trade coverage *explicitly*.

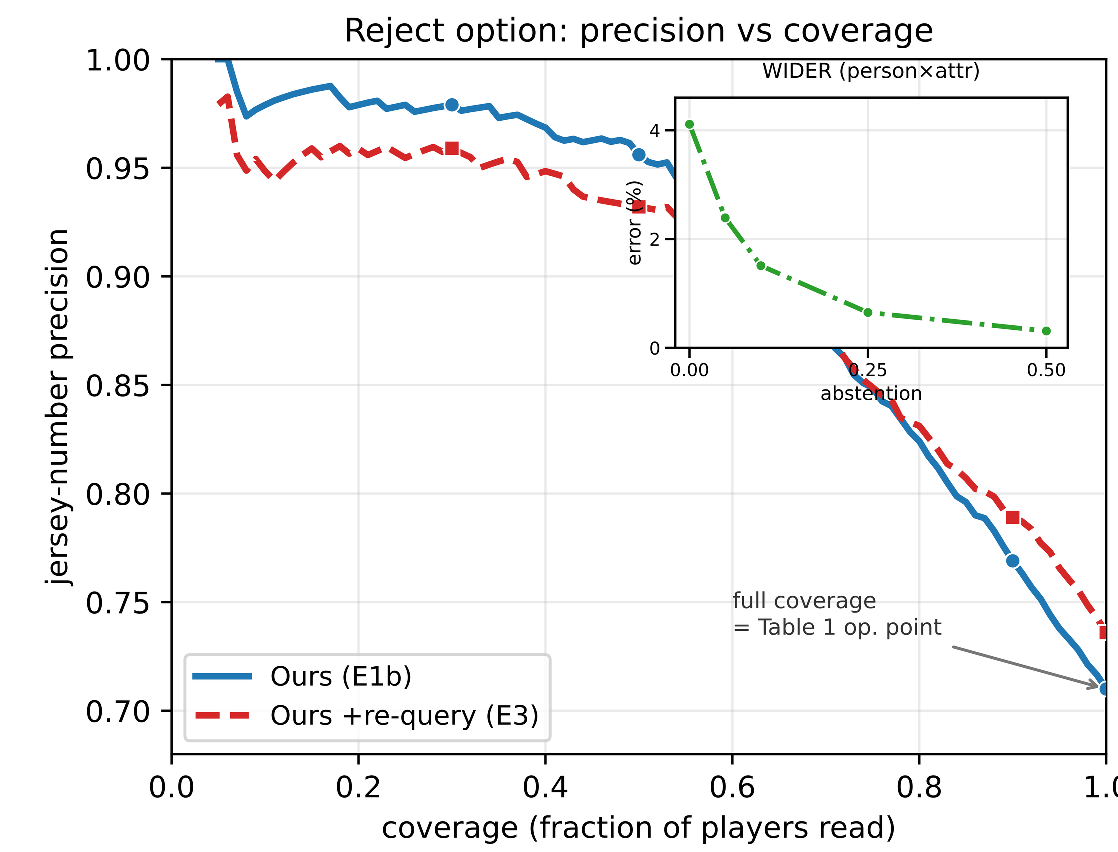
Training-free crop re-query

Small athletes are the dominant error mode. Players that are **small** (bottom quartile by box area) or **uncertain** (bottom quartile by *jersey_conf*) are re-read from a padded crop with the *same* model, same grammar capped at one player; re-reads replace pass-1 in place — boxes are never re-localised, so no duplicates.

tiny-bucket jersey F1	0.61 →**0.75**
AER cost (disclosed)	0.057 → 0.068
passes / image	1.93

Reject option: precision on demand

Ranking by *jersey_conf* and abstaining on the least-confident trades coverage for precision — a knob neither the cascades nor the closed APIs offer.



jersey precision **0.71** (full) → **0.96** (top half)

AUROC 0.879. Headline numbers are at *full* coverage — the reject option adds precision without quietly dropping recall.

It is a recipe, not a checkpoint

The identical recipe — same schema, grammar, frozen towers, confidence readout — transplanted to public **WIDER-Attribute**:

Setting	mAP
Prior specialists (a decade)	80.5–87.5
Ours, given-box	**93.1**
Ours, end-to-end	**84.5**

End-to-end = detect, crop and classify with *no* ground-truth boxes at any stage — to our knowledge the first detection-coupled result under this protocol.

The surprise: tuning is not the lever.

A LoRA ladder up to 3.15M trainable parameters moves the aggregate by <+0.0011 — no measurable gain under the configurations we test. The single *structural* choice of dropping quad supervision is worth +0.008, roughly 7× more.

In this regime the gains live in the **structure**, not in added weights.

Paper Code

Paper, supplementary & response: OpenReview, HCMIW Submission 14.

Code: eval suite, finite-state grammar, confidence recipe; sports numbers re-scorable from released caches.